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===== MASTER TUTOR SCRIPT: MHF4U UNIT 3

- RATIONAL FUNCTIONS Student: Rigden Total Slides: 40

HOW TO USE THIS SCRIPT: Keep this text file open on a second monitor or printed out. The HTML presentation file is 100% student-safe with no embedded teacher notes. Guide Rigden through the interactive tabs on each slide using these specific prompts.

SLIDE 1: Title Slide [SAY] Welcome back, Rigden! We've conquered polynomial functions, but what happens when we divide one polynomial by another? Welcome to Unit 3: Rational Functions. [SAY] This unit is all about asymptotes, holes, and fractions. Today's session is a 38-question gauntlet testing everything from basic restrictions to ultimate graphing challenges. Let's go!

LEVEL 1: WARM-UP (Slides 2 - 6)

SLIDE 2: Q1 - Domain Restrictions [ASK] What is the one rule in math regarding fractions? [EXPECT] You can never divide by zero. [SAY] Exactly. So factor the denominator and tell me what x cannot equal.

SLIDE 3: Q2 - VA vs. Hole [ASK] If a factor is in the denominator, it's usually a Vertical Asymptote. But when does it become a Hole (Point Discontinuity) instead? [EXPECT] When the exact same factor is also in the numerator and they cancel out. [SAY] Perfect. Factor the top and bottom of this equation and tell me what happens at $x = -4$.

SLIDE 4: Q3 - Horizontal Asymptote (Degree $<$) [SAY] Horizontal asymptotes are all about "End Behaviour" as x gets massive. [ASK] If the degree of the denominator is bigger than the numerator, who wins the tug-of-war as x approaches infinity? [EXPECT] The denominator gets bigger way faster, so the fraction approaches 0. HA is $y = 0$.

SLIDE 5: Q4 - Horizontal Asymptote (Degree $=$) [ASK] What happens when the degrees are tied? (Both are x^2). [EXPECT] You divide the leading coefficients. [SAY] Yes! So the HA is just $y = 5/3$.

SLIDE 6: Q5 - The Extraneous Root Trap [ASK] If the denominators are exactly the same, can we just set the numerators equal? [EXPECT] Yes, $x = 2$. [SAY] Wait! Did you check your restrictions from the original equation? [TIP] Emphasize that $x=2$ makes the original denominator 0, so it's an extraneous root. No solution!

LEVEL 2: MODERATE (Slides 7 - 14)

SLIDE 7: Q6 - Graph Intercepts [ASK] How do we find an x -intercept algebraically? (Set $y=0$, which means just setting the numerator to 0). [ASK] How do we find a y -intercept? (Set $x=0$).

SLIDE 8: Q7 - Hole Coordinates [SAY] We know the x -value of the hole is $x = 3$ because it cancels. [ASK] But a hole is a physical coordinate point. How do we find the y -value? [EXPECT] Substitute $x=3$ back into the SIMPLIFIED equation.

SLIDE 9: Q8 - End Behaviour Near VA [SAY] This is limit notation. As x approaches 2 from the positive side (2.01, 2.001). [ASK] If you plug 2.01 into $1/(x-2)$, do you get a massive positive number or massive negative number? [EXPECT] Positive. So it approaches positive infinity.

SLIDE 10: Q9 - Slant (Oblique) Asymptotes [ASK] We know HA rules for degree $<$ and degree $=$. What happens if the numerator degree is EXACTLY ONE greater than the denominator? [EXPECT] You get a slant (oblique) asymptote. [SAY] Correct! And how do we find its equation? (Polynomial Long Division or Synthetic Division).

SLIDE 11: Q10 - Finding the Slant Asymptote [SAY] Use synthetic division on this quadratic over linear. [ASK] Do we care about the remainder when finding a slant asymptote? [EXPECT] No, we drop the remainder. The quotient is the equation of the line $y = mx + b$.

SLIDE 12: Q11 - Basic Rational Inequality [SAY] To solve an inequality, we use a number line. We need the "critical values". [ASK] What are the two types of critical values for a rational function? [EXPECT] Roots (numerator = 0) and Vertical Asymptotes (denominator = 0).

SLIDE 13: Q12 - The Bounce Inequality [ASK] The factor $(x-3)$ is squared in the numerator. What does a squared factor do on a number line? [EXPECT] The sign bounces (stays the same) on both sides of that critical value, instead of alternating!

SLIDE 14: Q13 - Common Denominators [SAY] You want to solve $2/(x-1) + 3/(x+1) = 4/(x^2-1)$. [ASK] What is the LCD? [EXPECT] $(x-1)(x+1)$. Multiply every single term by this to clear the fractions instantly!

LEVEL 3: ADVANCED (Slides 15 - 23)

SLIDE 15: Q14 - Error Analysis: Inequalities [SAY] A student solved $(x-2)/(x+3) > 1$ by cross-multiplying the $(x+3)$. [ASK] Why is this a fatal error in inequalities? [EXPECT] We don't know if $(x+3)$ is positive or negative. If it's negative, the inequality sign was supposed to flip! [TIP] Rule: NEVER cross-multiply variables in an inequality. Bring the 1 over and find a common denominator.

SLIDE 16: Q15 - Subtracting over Inequalities [SAY] Let's do it the right way. Bring the x over so it's ≤ 0 . [ASK] What is the common denominator? Simplify the numerator to create a single fraction.

SLIDE 17: Q16 - Crossing the HA [ASK] Can a graph cross a Vertical Asymptote? (No). Can a graph cross a Horizontal Asymptote? (YES!) [SAY] To find where, find the HA (which is $y=2$), and then set the whole equation equal to 2 and solve for x .

SLIDE 18: Q17 - Word Problem: Work Rate [SAY] Classic Ontario exam question. Person A takes 4 hours, Person B takes 6. [ASK] What is their combined rate? [EXPECT] $1/4 + 1/6 = 1/t$. Solve for t !

SLIDE 19: Q18 - Word Problem: Mixtures [SAY] We are adding pure juice (100% or 1.0) to a 20% mix. [ASK] The volume increases by ' x ', and the pure juice increases by ' $1x$ '. Set the new fraction equal to 50% (0.5).

SLIDE 20: Q20 - Finding 'k' for a single VA [SAY] A quadratic denominator usually has TWO vertical asymptotes. [ASK] For it to only have ONE, the quadratic must be a perfect square. What discriminant rule from Grade 10 guarantees one solution? [EXPECT] $b^2 - 4ac = 0$.

SLIDE 21: Q21 - Working Backwards (a and b) [ASK] The HA is $y=3$. Look at ax/x . What does 'a' have to be? ($a=3$). [ASK] The VA is $x=-2$. Look at the denominator $x-b$. What does 'b' have to be? ($b=-2$).

SLIDE 22: Q22 - Even vs Odd Vertical Asymptotes [ASK] At $x=-3$, the graph shoots in opposite directions (one up, one down). At $x=2$, they both shoot UP like a volcano. [EXPECT] Opposite directions = odd degree on the VA factor (like degree 1). Same direction = even degree (squared).

SLIDE 23: Q23 - Word Problem: Average Cost [SAY] Average cost is Total Cost divided by x . $C(x)/x$. [ASK] What does the HA of the average cost function represent in the real world? [EXPECT] As production approaches infinity, the cost per item approaches the HA (the marginal variable cost).

LEVEL 4: CHALLENGE (Slides 24 - 31)

SLIDE 24: Q24 - Factoring Cubics for Holes [SAY] You have $(x^3 - 8)/(x - 2)$. [ASK] Use the Difference of Cubes formula to factor the top. Does $(x-2)$ cancel? (Yes, it's a hole, not a VA).

SLIDE 25: Q25 - Double VA Graphing Logic [ASK] If there are two VA's, the graph is split into 3 zones. The middle zone usually forms a U or an upside-down U. [SAY] Find the y-int and test a point to see if the U opens up or down.

SLIDE 26: Q26 - Tricky Rational Equation [SAY] $1/(x-1) = 1/(x+1)$. [ASK] If you cross multiply, you get $x+1 = x-1$, which simplifies to $1 = -1$. What does that mean? [EXPECT] No solution exists. The graphs never intersect.

SLIDE 27: Q27 - SA Remainder Interpretation [SAY] You found the SA is $y = x + 2$, and the remainder is 5. [ASK] Because the remainder is positive, is the curve approaching the asymptote from ABOVE or BELOW as x goes to positive infinity? [EXPECT] From above, because we are ADDING a tiny positive fraction to the line.

SLIDE 28: Q28 - Complex LCD Equation [SAY] The denominators are (x^2-9) , $(x-3)$, and $(x+3)$. [ASK] Notice anything about x^2-9 ? (It factors into the other two!). That makes the LCD very easy to find.

SLIDE 29: Q29 - Working Backwards (Holes and VA) [SAY] Hole at $x=2$ means $(x-2)$ on top and bottom. VA at $x=-1$ means $(x+1)$ on bottom. [ASK] Build the skeleton and plug in the point to find 'a'.

SLIDE 30: Q30 - Error Analysis: Cancellation [ASK] A student cancelled x from $(x^2+1)/x$ and got $x+1$. [EXPECT] Fatal algebra mistake. You can only cancel FACTORS (things being multiplied), never TERMS (things being added).

SLIDE 31: Q31 - Graphing Analysis [SAY] Looking at an equation to find Domain and Range. Domain is easy (exclude VA). [ASK] Range is harder. The HA is excluded, but are there any other holes excluding y-values?

LEVEL 5: MASTER CHALLENGE (Slides 32 - 38)

SLIDE 32: Q32 - Advanced Inequalities [SAY] $(x-2)(x+1) / (x-4)^2 \leq 0$. [ASK] The $(x-4)^2$ is always positive, so it doesn't flip signs. But x CANNOT equal 4. Mark it as an open circle on your number line!

SLIDE 33: Q33 - Slant Asymptote of Cubic/Quadratic [SAY] Slant asymptotes happen when the top is 1 degree higher. [ASK] Can you use synthetic division here? (No, divisor is degree 2). You MUST use full long division.

SLIDE 34: Q34 - Turning Points via Discriminant [SAY] Contest trick! Without calculus, find the turning point of $y = x / (x^2+1)$. [EXPECT] Rearrange to $yx^2 - x + y = 0$. Use $b^2 - 4ac \geq 0$ to find the max and min limits of y !

SLIDE 35: Q35 - Absolute Value in Rational [SAY] $|x-2| / (x-2)$. [ASK] If $x > 2$, what does this equal? (1). If $x < 2$, what does it equal? (-1). So it's a step function with a jump discontinuity!

SLIDE 36: Q36 - Oblique Asymptote Limits [SAY] This is prep for University Calculus. [ASK] As $x \rightarrow -\infty$, does the graph approach the slant asymptote from above or below?

SLIDE 37: Q37 - The Hidden Extraneous Root [SAY] Solve the equation algebraically. You get $x=3$ and $x=-2$. [ASK] Did you check the original denominators? $x=3$ creates a divide-by-zero error. Reject it.

SLIDE 38: Q38 - The Ultimate Boss Graph [SAY] Walk me through everything. Factored form, Domain, VA, Holes (with coordinates), HA, x and y intercepts, and test the zones.

SLIDE 39: Outro [SAY] 38 questions down. You have mastered Rational Functions. Review the Exam Traps list before your unit test. See you next class!